# FLAT PULLBACK OF A FAMILY IS THE COMPOSITION WITH THE RELATIVE CLASS OF THE FLAT MORPHISM

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file relative-cycles.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

*Allowed standard prerequisites (editorial).*

- Base change compatibility of the construction [X/X/Y]_e (relative-cycles chapter, Lemma lemma-family-associated-closed, canonical tag 0H4T). cited at proof lines 2895-2896. Identifies the fibre of the constructed family with the construction on the fibre.
- Flat pullback as the bivariant action of the relative class (relative-cycles chapter, Lemma lemma-flat-pullback-as-action, no canonical tag in the snapshot; no-tag pointer to the source label line). cited at proof lines 2896-2897. Supplies the fibrewise equality f_s^* beta_s = [X_s/X_s/Y_s]_e cap beta_s, which is the fibre of the claimed family identity.

*Source reference.* Original lemma 0H4T: sources/stacks-chow/tags.txt, source lines 21363–21363. Cited at proof lines 2895-2896: fibre of the constructed class is the construction on the fibre.

*Source reference.* Original lemma lemma-flat-pullback-as-action: sources/stacks-relative-cycles/pinned-original-source.tex, source lines 2448–2448. Cited at proof lines 2896-2897: supplies the fibrewise equality.

**Lemma 0.2.** *Let $f : X \to Y$ and $Y \to S$ be morphisms of schemes, both locally of finite type. Assume $f$ is flat of relative dimension $e$. Let $\beta$ be a family of $r$-cycles on the fibres of $Y/S$. Then $f^*\beta = [X/X/Y]_e \circ \beta$ as families of $(e + r)$-cycles on $X/S$.*

*Proof.* After unwinding the definitions and using that formation of $[X/X/Y]_e$ is compatible with base change (Lemma 0H4T) this follows from Lemma lemma-flat-pullback-as-action. □

*Date*: stacks-project:relative-cycles.tex:local-snapshot-20261012; GNU FDL 1.2 or later.